POZNAN UNIVERSITY OF TECHNOLOGY

Applying Statistical and Machine Learning Techniques for Customer Data Deduplication

Lessons Learned from an R&D Project in the Financial Sector

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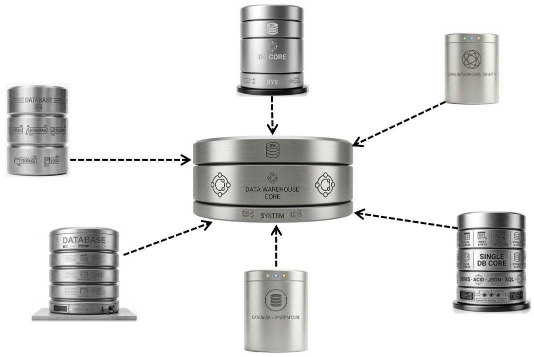
Outline

- **Fundamentals of data deduplication**
- **Our approach → statistical modeling**
- **ML/AI for data deduplication**
- **Our approach → ML/AI**
- **Summary, conclusions, and observations**
- **Based on the experience in cleaning and deduplicating customer records in the biggest Polish bank**
  - **3-years R&D project (2020-2023)**
  - **the developed method and software have been deployed in the production system**

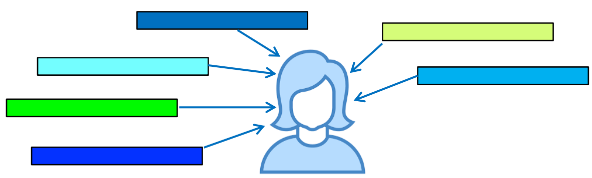
Context

- **Sources of duplicates (example)**
  - **separate financial products needs separate cust. records**
    - **accounts: savings/checking, brokerage, loan**
  - **company acquisitions**
    - **company X buys Y with all its customers**



Deduplication: introduction

- **deduplication = entity matching = entity resolution = entity reconciliation = duplicate identification = record linkage = reference reconciliation = data matching = string matching**
- **Discovering ALL groups of records that describe the same real world object (e.g., customer)**



these records are not identical

Deduplication: introduction

- **Before comparing, records should be cleaned (as much as possible)**
  - **homogenizing values (abbreviations, units of measurement, symbols, ...)**
  - **removing special signs, no punctuations**
  - **replacing abbreviations by full names**
- **Example: are these 2 records represent the same entity?**
  - **[Wrembel, Robert, ul. Matejki, Poznań]**
  - **[Wrębel, Robert, ul. Matejki, PonZan]**
- **Case 1: natural identifiers (e.g., national IDs) available**
  - **IDs must be real and error free → often unrealistic**
- **Case 2: no natural identifiers available**
  - **approximate/probabilistic decision based on record similarities**

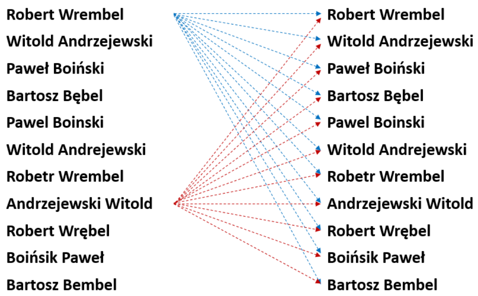
Deduplication: similarities

- **Computing similarities between pairs of records**
  - **similarities between attribute values → similarity measures**
  - **attribute importance → weights**

	TaxpayerID	Fname	Lname	Country	Title	Domain
	132018	Robert	Wrembel	Poland	Prof.	data wrehouses
sim: 0.7	w1	w2	w3	w4	w5	w6
	TaxpayerID	Fname	Lname	Country	Title	Domain
	132018	Robert	Wrębel	PL	professor	data integration

Deduplication: complexity

- **Deduplication needs comparing records**
  - **the worst case: each record compared to all the other records → the O(n^2) complexity**

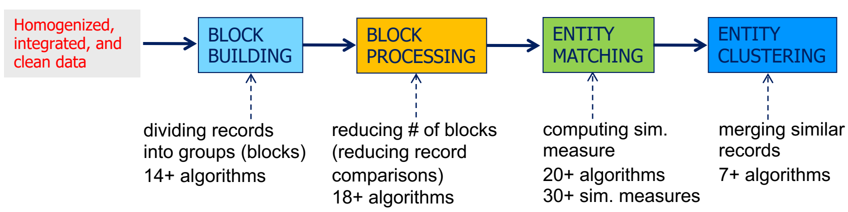


time for comparing values of 46char attribute, using Jaro-Winkler sim. measure (the fastest tested)

- **Bank example: over 20mln customer records**
  - **$2*10^7 * 2*10^7 = 4*10^{14}$ comparisons**
  - **one comparison = $3.6*10^{-7}$s → comparing the whole set: ~$14*10^7$s = 1700 days → must be optimized**

Deduplication pipeline

- **Improving performance**
  - **avoiding N^2 comparisons of records**
  - **finding groups of similar records faster**
- **State of the art base-line deduplication pipeline (BLDDP)**



- G. Papadakis, L. Tsekouras, E. Thanos, G. Giannakopoulos, T. Palpanas, M. Koubarakis: Domain- and Structure-Agnostic End-to-End Entity Resolution with JedAI. SIGMOD Record, Vol. 48, No. 4, 2019
- G. Papadakis, D. Skoutas, E. Thanos, T. Palpanas: Blocking and Filtering Techniques for Entity Resolution: A Survey. ACM Computing Surveys, 52:(3), 2021

Block building

BLOCK BUILDING → BLOCK PROCESSING → ENTITY MATCHING → ENTITY CLUSTERING

- **Goal: to eliminate not necessary comparisons of entities without discarding any matching entities**
- **Block building (a.k.a. blocking, indexing) organizes similar entities into the same block**
- **Comparisons of entities are made within the given block**
- **The right blocking key(s)?**
- **Popular algorithms**
  - **canopy clustering**
  - **hashing**
  - **token-based**
  - **sorted neighborhood**

Block building

BLOCK BUILDING → BLOCK PROCESSING → ENTITY MATCHING → ENTITY CLUSTERING

- **Sorted neighborhood**
  1. **sort records by a given attribute (list of attributes)**
     - **more similar records are located closer to each other**
  2. **compare records only within a moving window of N records**
     - **within the window all records compared with all records**

input data

RID	ULICA	MIASTO
1	F. RATAJCZAKA	Poznań
2	FR. RATAJCZAKA	Poznań
3	FR.RATAJCZAKA	Poznań
4	FRANCISZKA RATAJCZAKA	Gniezno
5	KĘPIŃSKA	Gniezno
6	KĘPIŃSKA	Gniezno
7	M PALACZA	Gniezno
8	M. PALACZA	Poznań
9	MACIEJA PALACZA	Pobiedziska
10	POWST WLKP	Pobiedziska
11	POWST. WLKP.	Poznań
12	POWST.WLKP	Poznań
13	POWSTANCOW WLKPIELKOPOLSKICH	Poznań
14	POWSTAŃCÓW WLKP.	Poznań
15	WARSZAWSKA 33	Poznań

- **Optimal window size?**

sorted neighborhood

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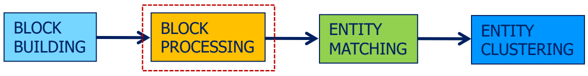
legend:
window size w=4 moving the window 1 rec. down comparing a rec. with previous rec. moving window

Block processing



- **Goal: to minimize the number of comparisons, i.e., to discard unnecessary comparisons:**
  - **between records in a given block**
  - **between blocks (in case of overlapping blocks)**
- **A popular algorithm: meta blocking [1, 2]**

1. G. Papadakis, G. Koutrika, T. Palpanas, W. Nejdl: Meta-Blocking: Taking Entity Resolution to the Next Level. IEEE Trans. Knowl. Data Eng. 26(8), 2014
2. F. Zhang, Z. Gao, K. Niu: Pruning algorithm for Meta-blocking based on cumulative weight. Journal of Physics: Conference Series, 887, 2017

Block processing: meta blocking

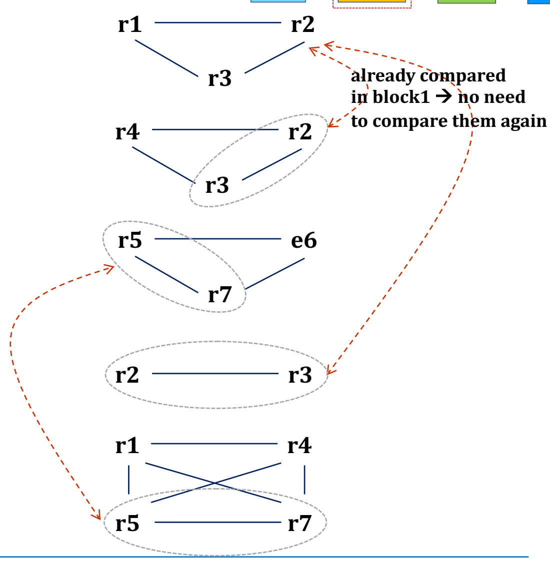
block1: {r1, r2, r3}

block2: {r2, r3, r4}

block3: {r5, r6, r7}

block4: {r2, r3 }

block5: {r1, r4, r5, r7}

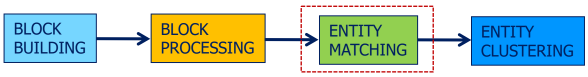


Block processing

- Other algorithms
  - block filtering [1]
  - splitting large blocks [2]
  - merging blocks with similar blocking keys [3]
  - size-based block clustering [4]
  - iterative blocking [5]
  - block pruning + duplicate propagation [6]

1. G. Papadakis, G. Papastefanatos, T. Palpanas, M. Koubarakis: Scaling Entity Resolution to Large, Heterogeneous Data with Enhanced Meta-blocking. EDBT, 2016
2. A. Das Sarma, A. Jain, A. Machanavajjhala, P. Bohannon: An automatic blocking mechanism for large-scale de-duplication tasks. CIKM, 2012
3. J. Fisher, P. Christen, Q. Wang, E. Rahm: A Clustering-Based Framework to Control Block Sizes for Entity Resolution. KDD, 2015
4. J. Fisher, P. Christen, Q. Wang, E. Rahm: A Clustering-Based Framework to Control Block Sizes for Entity Resolution. KDD, 2015
5. S.E. Whang, D. Menestrina, G. Koutrika, M. Theobald, H. Garcia-Molina: Entity resolution with iterative blocking. SIGMOD, 2009
6. G. Papadakis, E. Ioannou, C. Niederée, P. Fankhauser: Efficient entity resolution for large heterogeneous information spaces. WSDM, 2011

Entity matching

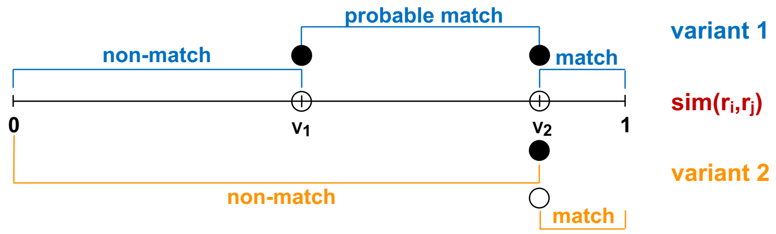


- Goal: to assign a label to each pair of entities being compared
  - variant 1: match / probable match / non-match
  - variant 2: match / non-match

- M. Alamuri, B.R. Surampudi, A. Negi: A survey of distance/similarity measures for categorical data. Int. Joint Conf. on Neural Networks (IJCNN), IEEE, 2014
- S. Boriah, V. Chandola, V. Kumar: Similarity Measures for Categorical Data: A Comparative Evaluation. SIAM Int. Conf. on Data Mining (SDM), 2008
- P. Christen. 2012. Data Matching - Concepts and Techniques for Record Linkage, Entity Resolution, and Duplicate Detection. Springer, 2012
- P. Christen: A Comparison of Personal Name Matching: Techniques and Practical Issues. Int. Conf. on Data Mining (ICDM), IEEE, 2006
- F. Naumann: Similarity measures. Hasso Plattner Institut, 2013
- M. del Pilar Angeles, A. Espino-Gamez. 2015. Comparison of Methods Hamming Distance, Jaro, and Monge-Elkan. Int. Conf. on Advances in Databases, Knowledge, and Data Applications, 2015

Entity matching

- Matching uses **similarity function** (measure) $sim(r_i,r_j)$ that maps each pair of records (r_i,r_j) to a similarity value



Similarity measures for text data

- Over 30 different similarity measures for text data
- **Phonetic encoding** of a text value
  - e.g., Soundex, Phonex, Phonix, NYSIIS, Metaphone
- **Edit distance** - count the smallest number of edit operations that are required to convert string s1 into s2
  - e.g., Levenshtein, Damerau-Levenshtein, Smith-Waterman
- **N-grams** - split compared strings into n-character sub-strings (n-grams) and count common n-grams
- **Set similarity** - a similarity depends on the number of common elements in compared sets
  - e.g., Overlap, Jaccard, Sorensen-dice

Similarity measures for text data

- The longest common sub-sequence or the longest common sub-string
- Vector representation in an m-dimensional space
  - e.g., TFIDF, Cosine
- Compression techniques
  - e.g., BZ2, Lempel-Ziv-Markov, ZLib
- Ensemble of measures
  - e.g., Monge-Elkan + Damerau-Levenshtein + n-gram, Cosine+n-gram
- **Which similarity measure is adequate for a given problem?**

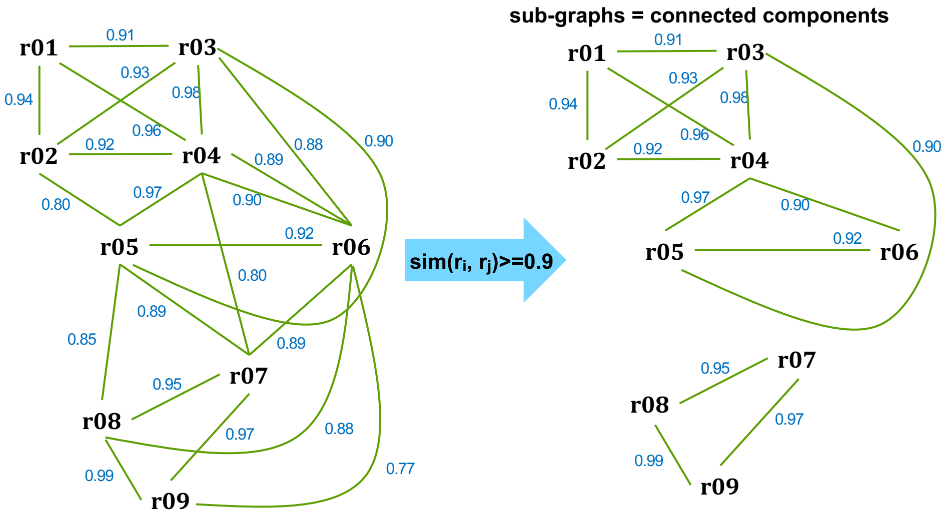
BLOCK BUILDING → BLOCK PROCESSING → ENTITY MATCHING → ENTITY CLUSTERING

Entity clustering

BLOCK BUILDING → BLOCK PROCESSING → ENTITY MATCHING → ENTITY CLUSTERING

- Duplicates may concern more than 2 records
- Compared pairs of records with their similarities are merged into graphs
  - nodes: records
  - edges: similarity values
- Goal: find sub-graphs (clusters) of records that represent the same real-world object (with a probability > threshold)

Entity clustering



Entity clustering

- **Popular clustering algorithms (from the research on graphs)**
  - **maximum clique, e.g., Born-Kerbosh [1]**
    - **too strict as cliques are rare**
  - **Highly Connected Sub-Graphs [2, 3]**
  - **Ricochet [4]**

1. C. Bron, J. Kerbosch: Finding all cliques of an undirected graph (algorithm 457). Communications of the ACM 16(9), 1973
2. E. Hartuv, R. Shamir: A clustering algorithm based on graph connectivity. Information Processing Letters 76(4-6), 2000
3. F. Hüffner, C. Komusiewicz, A. Liebtrau, R. Niedermeier: Partitioning biological networks into highly connected clusters with maximum edge coverage. IEEE/ACM Transactions on Computational Biology and Bioinformatics 11(3), 2014
4. D.T. Wijaya, S. Bressan: Ricochet: A Family of Unconstrained Algorithms for Graph Clustering. DASFAA, 2009

BLDDP: challenges

BLOCK BUILDING → BLOCK PROCESSING → ENTITY MATCHING → ENTITY CLUSTERING

14+ algorithms | 18+ algorithms | 20+ algorithms | 7+ algorithms

- **C1: which algorithms to apply to a given data set? → selecting the best combination of the algorithms**
  - **deduplication quality (precision, recall)**
  - **execution costs**
  - **35280+ combinations of these algorithms**
  - **searching the full space of alg. combinations is impossible**
  - **some methods may be better for deduplicating personal data, some for bibliographical data, some for addresses**
    - **knowledge of a domain expert is crucial**
  - **an automatic approach does not exist**

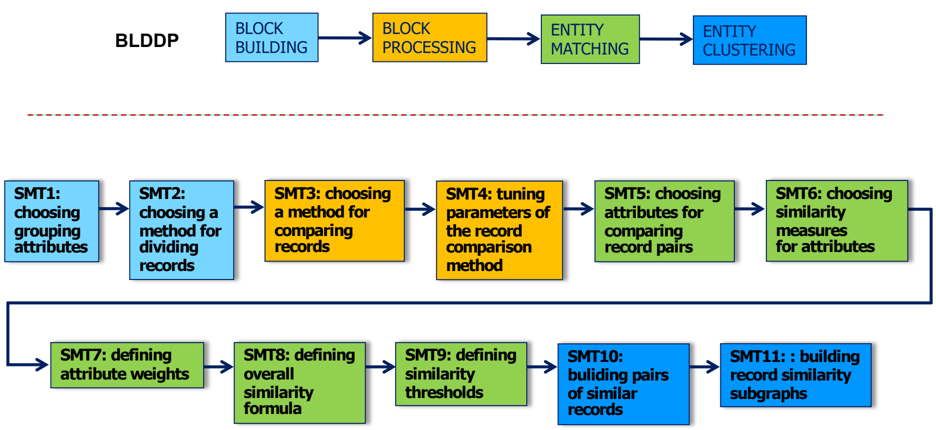
BLDDP: challenges

- **C2: blocking key(s)**
- **C3: window size in sorted neighborhood**
- **C4: adequate attributes for computing similarity**
- **C5: adequate similarity measures**
- **C6: weights of attributes being compared**
- **C7: similarity thresholds between classes: duplicates, probably duplicates, not duplicates**

Our approach

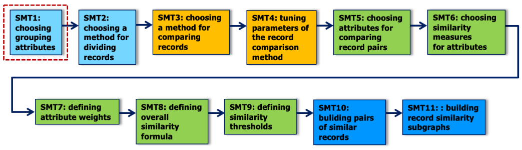
- **Data quality**
- **Input data are partially dirty**
- **Why partially dirty data?**
  - **customer first and last names cannot be cleaned without a customer permission (by law)**
  - **not all data can be cleaned automatically**
  - **too many records need manual cleaning → impossible to clean 20+ million of customer records within a limited project time**

Our DDP



- **SMT1-SMT11 address challenges C1-C7**

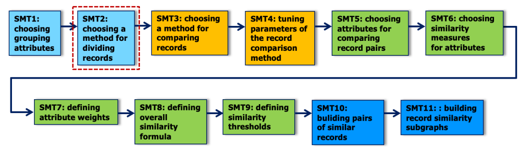
SMT1



- **Extension of [1] to handle**
  - **a diversity of values of each attribute → modelled by means of the Gini Index**
  - **the size of a record group → penalized by means of a quadratic function**
- **Observations**
  - **the formula to rank attributes → easy to understand and to compute**
  - **the produced ranking of attributes verified and approved by domain exerts**

1. L. de Souza Silva, F. Murai, A.P.C. da Silva, M.M. Moro, M.M.: Automatic identification of best attributes for indexing in data deduplication. A. Mendelzon Int. Workshop on Foundations of Data Management, CEUR Workshop Proceedings, vol. 2100, 2018

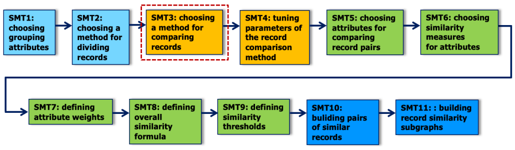
SMT2



- **Sorting attributes - requirements**
  - **collocate duplicates close to each other**
  - **do not include nulls (if possible)**
  - **include low number of erroneous values (if possible)**
  - **are immutable**
- **The final sorting attributes were selected based on**
  - **data profiling**
  - **a statistical method that we developed for this purpose (SMT1)**
  - **expert knowledge**
  - **multiple experiments with various attribute combinations**

SMT3

- **Record comparison → sorted neighborhood**



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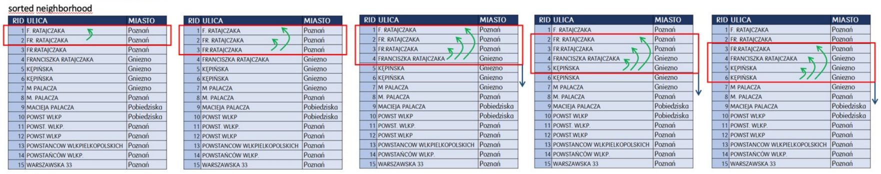
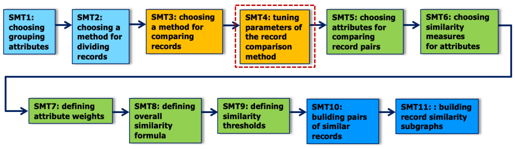
sorted neighborhood



legend:
window size w=4 moving the window 1 rec. down comparing a rec. with previous rec. moving window

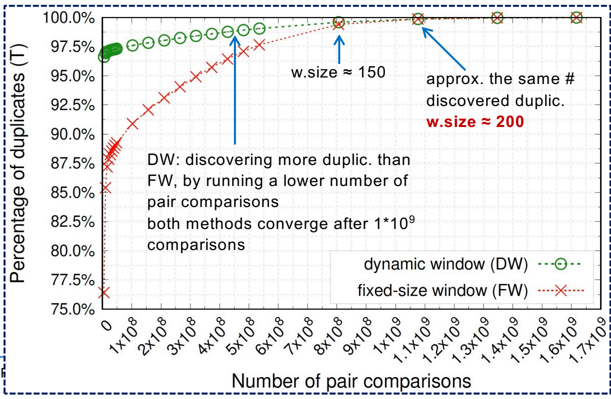
SMT4

- **Window size**
  - **bigger window → more records to compare → processing time**
  - **bigger window → more duplicates discovered**

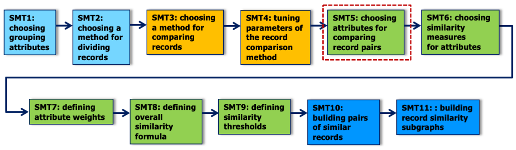


- **fixed size window → optimal size?**
- **dynamic window → max size?**

% expressed w.r.t. max # duplicates possible to discover for the window of size 300

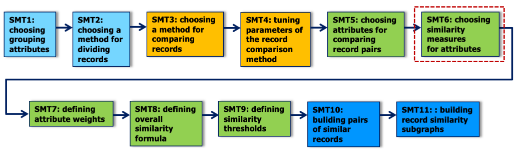


SMT5



- **Candidates for comparison**
  - **natural record identifiers (if available and correct)**
  - **without nulls (if possible)**
  - **immutable**
  - **with cleaned and homogenized values (if possible)**
- **Few attributes have these features in reality**
- **Attributes were selected based also on expert knowledge**

SMT6

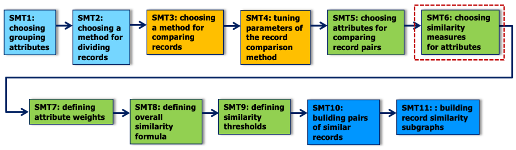


compared strings
- Avenida de la Reina María Cristina
- ave. de la reina M.Cristina

sim.measure	Python package	similarity value
Levenshtein	strsimpy	0.676
Jaro	jellyfish	0.740
Jaccard	textdistance	0.694
1-gram	ngram	0.694
2-gram	ngram	0.465
Overlap	textdistance	0.926

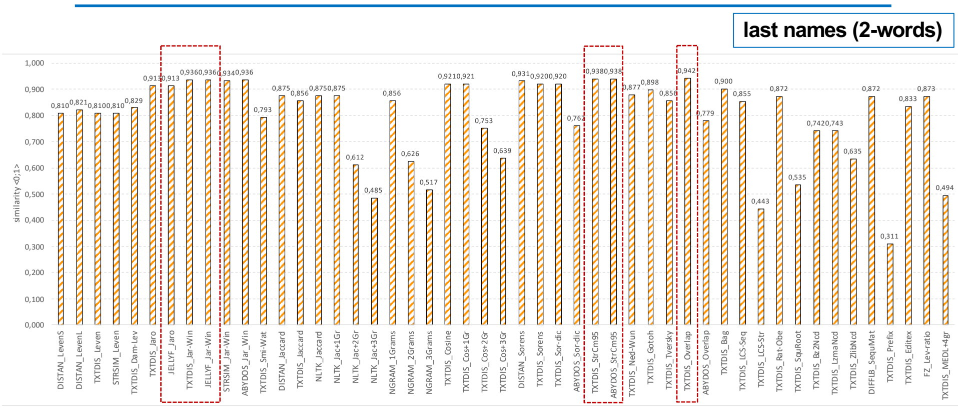
- **Comment 1: for the same compared strings, different similarity measures return different values**
- **Comment 2: the same measure, but implemented in different packages, may return different values**

SMT6: experiments



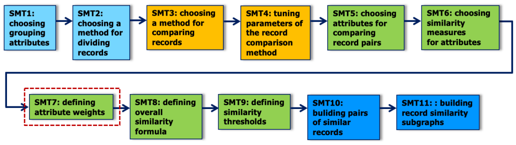
- **The evaluation of 26 basic, 6 simple ensemble, and 77 complex ensemble similarity measures available in Python packages**
  - **distance, textdistance, strsimpy, jellyfish, nltk, ngram, difflib, fuzzywuzzy**
- **Experiments run on a real data sets**
- **All test data represented true positives, but with typical real errors found in customers data**

SMT6: experiments



- **Based on our findings, in the project, we applied:**
  - **Jaro-Winkler for variable length texts**
    - **other promising measures: StrCmp95 and Overlap**
  - **Hamming for fixed length texts like birthdates and national IDs**

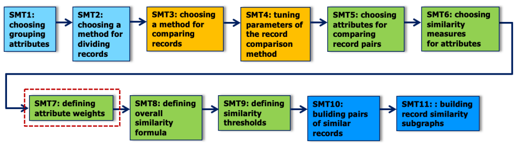
SMT7



Notice: in practice, SMT7, SMT8, and SMT9 are carried out simultaneously

- **Some attributes are more important than others, e.g.:**
  - **birth date – not unique, but distinctive**
  - **first name – not unique, usually not distinctive**
- **Introducing attribute weights → weighted similarity for pairs of records**
- **Problem: defining weights for attributes**

SMT7

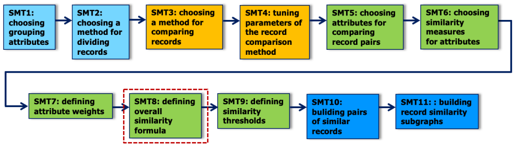


- **Training data set needed**
  - **1000 of pairs labeled by experts**
  - **hand-crafted rules induced from the labeled pairs**

```
IF X THEN Non-duplicate
IF X THEN Probable duplicate
IF X THEN True duplicate
```

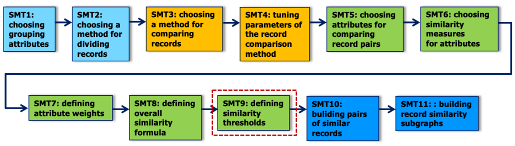
  - **X is a complex condition**
  - **the rules were used for labeling remaining records**

SMT8



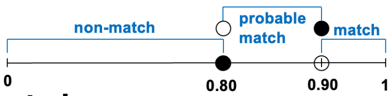
- **Hand-crafted similarity formula verified by Bank experts**
- **Weighted similarity of the whole records in a pair:**
  - **null values problem**
    - **skip attributes with null values in similarity calculation**
    - **skip comparison if there are too many missing values**
  - **technical values**
    - **national ID vs. artificial national ID – skip attribute**
  - **domain expert knowledge**

SMT9

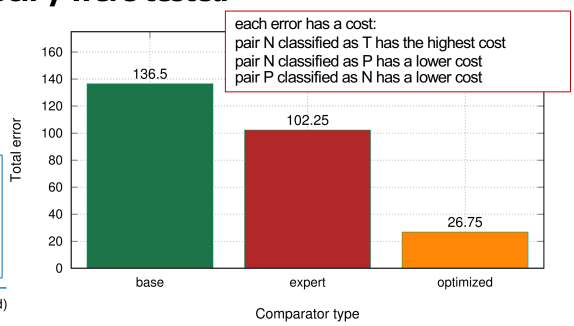


- **Mathematical programming**
- **Problem: minimize a function that computes a total misclassification cost on all expert-labelled pairs**

final similarity thresholds



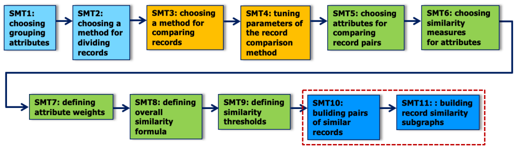
- **Three algorithms from SciPy were tested**
  - **Nelder-Mead**
  - **Powell (the best)**
  - **COBYLA**



base – all weights eq. 1
expert – weights tuned with the help of the experts
optimized – using mathematical programming

SMT10 and SMT11

- Sorted neighborhood produces pairs of similar records

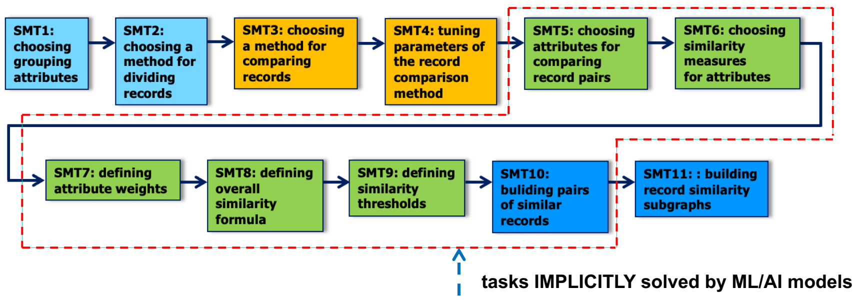


- Duplicates may concern more than 2 records
  - record similarity graph must be constructed
  - connected components represents groups of similar records
- Modified HCS algorithm
  - original HCS [1]
  - modified HCS [2]

1. E. Hartuv, R. Shamir: A clustering algorithm based on graph connectivity. Information Processing Letters, 76 (4–6), 2000, doi:10.1016/S0020-0190(00)00142-3
2. W. Andrzejewski, B. Bębel, P. Boiński, R. Wrembel: On tuning parameters guiding similarity computations in a data deduplication pipeline for customers records: Experience from a R&D project. Information Systems, vol. 121, 2024

ML for Deduplication

- ML/AI for data deduplication
- Tasks in the deduplication pipelines are complex → apply ML/AI to solve them implicitly



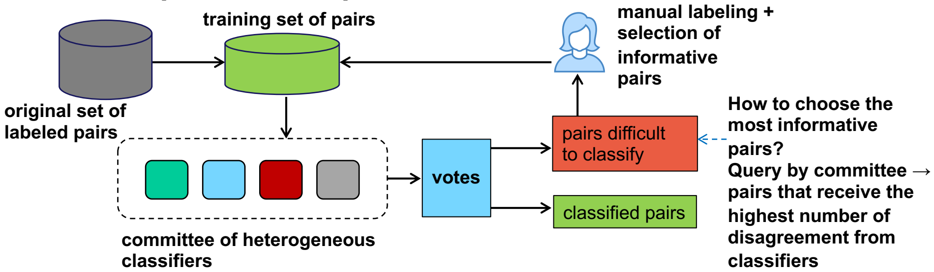
- Standard solution → classification
  - classes: T, N
  - classes: T, P, N

ML for deduplication: challenges

- **C8: how to construct a learning data set of a reasonable size for 20M customers' database?**
  - **what is the minimum size of this set?**
  - **manual labeling of 1000 pairs of rows**
    - **3 persons, including an expert from the FI**
    - **labeling throughput: 1.33 rec/min**
    - **22.2h for labeling 1000 rows**
  - **conclusion: impossible to construct a learning data set manually**
- **C9: how to efficiently verify a model on hundreds of thousands of records? → pertinent to:**
  - **standard ML for deduplication**
  - **deep learning for deduplication**
- **C10: explainability of AI deduplication models (neural networks)**

ML for deduplication: Active Learning

- **Alternative solution → Active Learning**
  - **at each iteration, pairs difficult to classify (the most informative) are labeled manually by experts**
  - **the labeled pairs are added into the training set for the next iteration of retraining a classifier**
  - **repeat until a stop condition is met**



- **Self Learning**
  - **correctly classified pairs extend the training set**

Active Learning

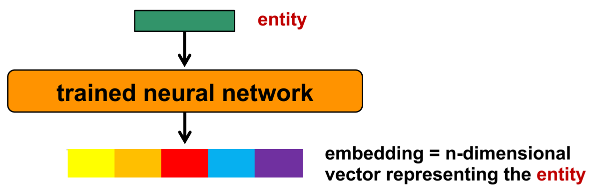
- **Frequently used classifiers**
  - **decision tree**
  - **support vector machines**
  - **logistic regression**
  - **one-vs-rest logistic regression**

- **Example approaches**

- S. Tejada, C.A. Knoblock, S. Minton: Learning object identification rules for information integration. Inf. Syst. 26, 2001
- S. Sarawagi, A. Bhamidipaty: Interactive deduplication using active learning. SIGKDD, 2002
- Q. Wang, D. Vatsalan, P. Christen: Efficient Interactive Training Selection for Large-Scale Entity Resolution. PAKDD, 2015
- X. Chen, Y. Xu, D. Broneske, G. C. Durand, R.Zoun, G. Saake: Heterogeneous Committee-Based Active Learning for Entity Resolution (HeALER). ADBIS 2019
- A. Jurek, J. Hong, Y. Chi, W. Liu: A novel ensemble learning approach to unsupervised record linkage. Inf. Syst. 71, 2017

Embedding

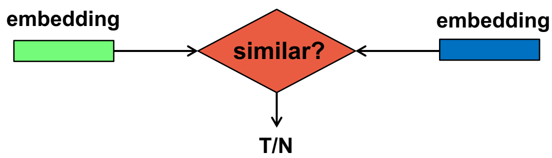
- **Trained neural network transforms an entity it into an n-dimensional vector representation**



- **The embeddings of semantically related entities are close to each other in a semantic space**

Deep Learning for deduplication

- **Key idea: treat matching as binary classification problem using entities embeddings**
  - **similarity between two entities → distance between their corresponding embeddings**
- **Blocking phase:**
  1. **find an embedding for each entity**
  2. **find candidate pairs (e.g., using K-nearest neighbors)**
- **Matching phase:**
  1. **embeddings of candidate pairs are the input of another binary classifier**
  2. **the classifier decides if the candidate pair is a match or not**



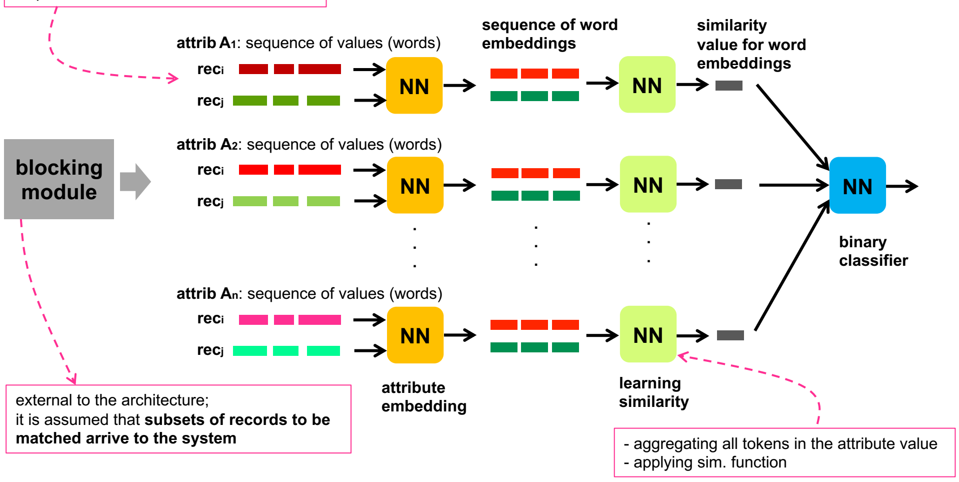
DL for deduplication

- **Example solutions**

1. M. Ebraheem, et al.: Distributed Representations of Tuples for Entity Resolution. PVLDB 11, 11, 2018
2. W. Zhang, et al.: AutoBlock: A hands-off blocking framework for entity matching. WSDM, 2020
3. S. Thirumuruganathan et al.: Deep Learning for Blocking in Entity Matching: A Design Space Exploration. VLDB Endow., 2021
4. S. Mudgal, et al.: Deep Learning for Entity Matching: A Design Space Exploration. SIGMOD, 2018
5. B.Li, et al.: GraphER: Token-Centric Entity Resolution with Graph Convolutional Neural Networks. IAAI, 2020
6. A. Vaswani, et al.: Attention Is All You Need. NIPS, 2017
7. J. Devlin, M. Chang, K. Lee, K. Toutanova: BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. Proc. of NAACL-HLT, 2019
8. U. Brunner, K. Stockinger: Entity Matching with Transformer Architectures - A Step Forward in Data Integration. EDBT, 2020
9. R. Chen, Y. Shen, D. Zhang: GNEM: A Generic One-to-Set Neural Entity Matching Framework. WWW, 2020
10. Y. Li, J. Li, Y. Suhara, A. Doan, W. Tan: Deep Entity Matching with Pre-Trained Language Models. PVLDB, 2020
11. R. Peeters, C. Bizer: Dual-Objective Fine-Tuning of BERT for Entity Matching. PVLDB, 2021
12. A. Zeakis, G. Papadakis, D. Skoutas, M. Koubarakis: Pre-trained Embeddings for Entity Resolution: An Experimental Analysis. PVLDB, 2023
13. A. Jain, S. Sarawagi, P. Sen: Deep Indexed Active Learning for Matching Heterogeneous Entity Representations. VLDB Endow. 15(1), 2021

DeepMatcher

- S. Mudgal, H. Li, T. Rekatsinas, A. Doan, Y.Park, et al.: Deep Learning for Entity Matching: A Design Space Exploration. SIGMOD, 2018



LLM for deduplication?

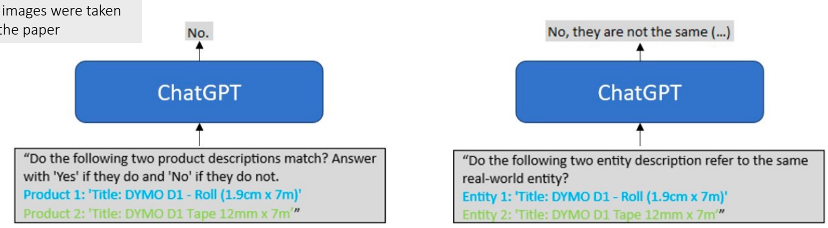
- "**LLM-based** entity matching methods can achieve **similar or even better performance** than deep learning methods trained on large amounts of data, and are less susceptible to the unseen entity problem"

T. Wang, et. al.: Match, Compare, or Select? An Investigation of Large Language Models for Entity Matching. arXiv preprint arXiv:2405.16884, 2024

LLM for EM (appr. 1)

- **Focus: testing performance of various prompt designs**
  - **simple language and answer**
  - **complex language and answer**
  - **model answer**
    - **free text or**
    - **forced to yes/no**
  - **#attributes compared in pair of records: 1, 2, 3**

these images were taken from the paper



- R. Peeters, C. Bizer: Using ChatGPT for Entity Matching. ADBIS (Short Papers), 2023

LLM for EM (appr. 1)

- **Adding rules to prompt**
  - **F1 increases by 6%**

Task Desc.	Your task is to decide if two product descriptions match. The following rules regarding product features need to be observed:
Rules	1. The brand of matching products must be the same if available 2. Model names of matching products must be the same if available 3. Model numbers of matching products must be the same if available 4. Additional features of matching products must be the same if available
Task Desc.	Do the following two product descriptions match? Answer with 'Yes' if they do and 'No' if they do not.
Task Input	Product 1: 'Title: DYMO D1 – Roll (1.9cm x 7m)' Product 2: 'Title: DYMO D1 Tape 12mm x 7m'

this image was taken from the paper

- **Real case (our project)**
  - **89 rules with different voting power → performance**

LLM for EM (appr. 2)

- Extension of appr. 1
- Explaining decisions
  - importance of attributes
  - similarity values

Actor	Prompt (Walmart-Amazon)
User	"Do the two entity descriptions refer to the same real-world entity? Answer with 'Yes' if they do and 'No' if they do not. **Entity 1:** night owl night owl zeus-810 16-channel h. 264 dvr kit with 1tb hard drive 773.82 **Entity 2:** night owl optics security zeus-810 16-channel 8-camera h. 264 dvr surveillance kit with d1 recording and hdmi output zeus-810 676.63"
AI	Yes.
User	Now **explain your decision in a structured format**, listing the attributes that you compared for reaching your decision. Each attribute should be accompanied by the attribute values and a score between -1 and 1 that shows the **importance of the attribute** for the decision. If the attribute influenced the decision towards non-match the importance score should be negative. If the attribute pointed towards a match, the importance score should be positive. Also provide a **similarity score for the attribute values**. If an attribute only occurs in one item, specify the value of that attribute for the other item as "missing". An example output is the following: attribute=brand - importance=0.05 - values=Logitech###Logitech - similarity=1.00 attribute=model - importance=-0.95 - values=MX G500###MX Master 3S - similarity=0.20 attribute=color - importance=0.00 - values=missing###Graphite - similarity=0.00'
AI	**attribute=brand** - importance=0.30 - values=Night Owl###Night Owl Optics - similarity=0.90 **attribute=model** - importance=0.40 - values=Zeus-810 16-channel H. 264 DVR kit###Zeus-810 16-channel 8-camera H. 264 DVR surveillance kit - similarity=0.85 **attribute=price** - importance=0.20 - values=773.82###676.63 - similarity=0.80 **attribute=hard drive** - importance=-0.10 - values=1TB###missing - similarity=0.00

this image was taken from the paper

- R. Peeters, A. Steiner, C. Bizer: Entity Matching using Large Language Models. EDBT, 2026

LLM for EM (appr. 2)

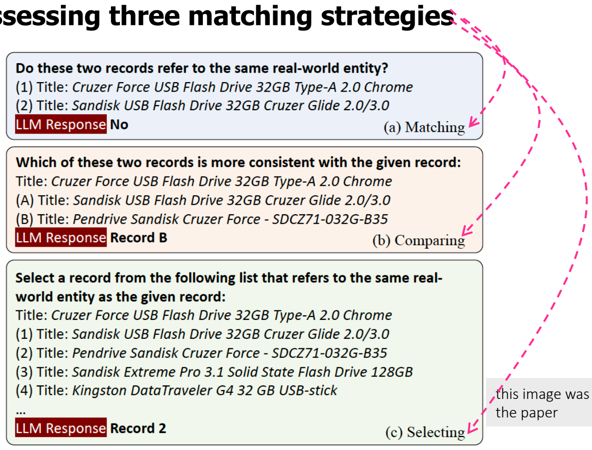
- Error analysis

Actor	Prompt
User	The following list contains false positive and false negative product pairs from the output of a product matching classification system. **Given the product pairs and the associated explanations, come up with a set of error classes**, separately for both false positives and false negatives, **that explain why the classification systems fails on these examples.** **False Negatives:** FN1 {Entity 1} {Entity 2} {Explanation} {additional FNs} **False Positives:** FP1 {Entity 1} {Entity 2} {Explanation} {additional FPs}
AI	Based on the provided examples of false negatives (FN) and false positives (FP), we can identify several error classes that explain why the classification system fails on these examples. **### Error Classes for False Negatives:** 1. **Model Number Mismatch**: The system fails when there are slight differences in model numbers or product codes, even when other attributes match closely. 2. **{additional error classes}** **### Error Classes for False Positives:** 1. **Overemphasis on Matching Attributes**: The system might give too much weight to matching attributes like brand or model number, leading to false positives even when other important attributes differ. 2. **{additional error classes}**

this image was taken from the paper

LLM for EM (appr. 3)

- Focus on assessing three matching strategies



this image was taken from the paper

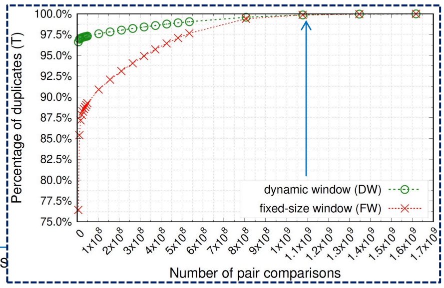
- T. Wang, et. al.: Match, Compare, or Select? An Investigation of Large Language Models for Entity Matching. arXiv preprint arXiv:2405.16884, 2024

LLM: research vs. reality

- Experimental evaluation in research
  - Y. Li, J. Li, Y. Suhara, A. Doan, and W. Tan. Deep entity matching with pre-trained language models. Proc. VLDB Endowment, 14(1), 2020
    - two tables with 789K and 412K of rows
  - R. Peeters, C. Bizer: Using ChatGPT for Entity Matching. ADBIS (Short Papers), 2023
    - ~40K of rows
  - A. Primpeli, C. Bizer: Profiling Entity Matching Benchmark Tasks. CIKM, 2020
    - ~ 50K of rows
- No time performance evaluation

LLM: research vs. reality

- **R. Peeters, C. Bizer: Using ChatGPT for Entity Matching. ADBIS (Short Papers), 2023**
  - **cost: 0.14 eurocent/comparison**
  - **time: 1.69 h for comparing $10*10^6$ candidate pairs (top 20 rows compared)**
- **Possible costs in a real (our) project**
  - **#rows in a data set for deduplication: $20*10^6$**
  - **#pair comparisons: $1.1*10^9$**
    - **sorted neighborhood: dynamic window or fixed size window of 200 rows**
  - **cost: 0.14 eurocent/comparison**
  - **total cost: 154 000 000 EUR**
  - **total time: 186h (7.7 days)**
- **Our process takes ~20h**



Our ML deduplication pipeline

MLT1: record labelling → **MLT2:** building labeling rules → **MLT3:** record labeling by rules → **MLT4:** building learning set and test set → **MLT5:** feature engineering → **MLT6:** model building (classifier) and tuning → **MLT7:** model testing and tuning

Optuna for hyperparam. tuning

Classifiers
- decision trees
- random forest
- SVM
- FF neural network

MLT1 and MLT2

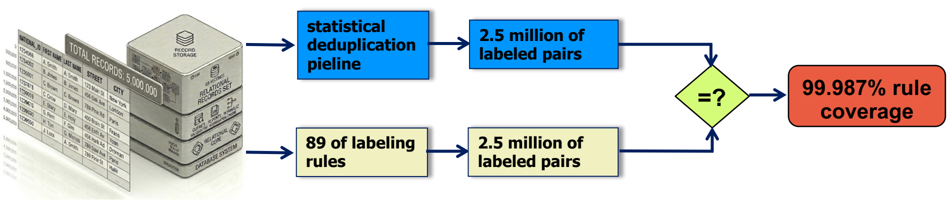
MLT1: record labelling → **MLT2:** building labeling rules

- **Labeling pairs**
  - **a dataset with 1000 pairs labeled by experts**
  - **89 rules involving 18 attributes built from this dataset**
  - **class labels**
    - **non-duplicates - N**
    - **true duplicates - T**
    - **probable duplicates - P**

MLT3

MLT3: record labeling by rules

- **To verify the coverage of the rules we used 2.5 milion pairs created from 5 milion records by the statistical pipeline**
- **Out of 2.5 milion pairs only 23352 pairs were not covered by any rule → rule coverage: 99.987%**



MLT4

MLT4: building learning set and test set

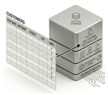
- **Rule-labelled pairs created in MLT3 were used to create a training and testing set**
  - **50000 pairs were randomly sampled and divided into training and testing data sets 50/50**

89 of labeling rules → 2.5 million of labeled pairs

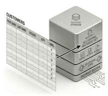
selecting pairs by stratified sampling

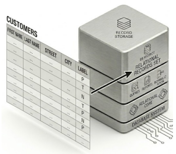
split: 0.5

25000 of training pairs



25000 of testing pairs





MLT6 and MLT7

MLT6: model building (classifier) and tuning → MLT7: model testing and tuning

- **Classification models tested (from sklearn)**
  - **decision tree (DecTree)**
  - **random forest (RandFor)**
  - **SVM**
  - **feed forward neural network (FFNN)**
- **Hyperparameters were adjusted**
  - **manually for DecTree and RandFor**
  - **via optuna for SVM and FFNN**
- **False negatives preverred over false positives**
  - **precision > recall**

	DecTree	RandFor	FFNN	SVM	StatModel
Precision	0.9577	0.9629	0.9444	0.8898	0.8051
Recall	0.9399	0.9551	0.9368	0.9619	0.8182

Summary

- R&D deduplication project for the biggest Polish Bank
- The statistical and ML pipelines were implemented
- The statistical pipeline was deployed in the production system
  - it is running deduplicating batches of 20+ mln of customer records

Challenges of real R&D projects

- The quality of data being deduplicated → far from perfect
- The lack of labeled data for ML algorithms
- Data volume
- Limited time for a project
- Limited monetary budget for a project
- Limited development environment → only certified software
- Solutions
  - the simpler the better
  - intuitive/understandable
  - explainable
  - easy to extend and customize

Challenges of real R&D projects

- **The solution must be deployable in the IT architecture of the Bank (specific hardware and software)**
- **The developed methods must be efficient → they will be applied to millions of rows**

- **For large data repositories it is impossible to discover ALL duplicates**
  - **a realistic case: to discover as many duplicates as possible**